Map of data collection areas in Madagascar

2024-02-07

```
res_dir <- "../results/s09_data_source_map/"
```

Create a map showing data collection locations in Madagascar

```
target <- c("Madagascar", "Mozambique", "Comoros", "Mauritius", "Seychelles", "Mayotte", "United Republi

# @TODO "Reunion" "Equatorial Guinea"!!
madagascar_inset <- ne_countries(
  country = target,
  type = "map_units",
  scale = 50
)
```

```
# Dissolve map country boundaries into a single geometry
madagascar_dissolved <- madagascar_inset %>%
  st_union() %>%
  st_as_sf() # Convert back to sf object
```

```
# Africa
africa <- ne_countries(
  continent = "Africa",
  scale = 50,
  returnclass = "sf"
) %>%
  st_union() %>%
  st_as_sf()
```

Read in sampling locations

```
kml_data <-
  readRDS("../data/raw/data_input/sampling_locations_with_pierron.rds")

# add a column to indicate whether they are from our sample or
# from the literature
ours <- rep(0, 5)
pierron <- rep(1, 9)
sample <- c(ours, pierron)
kml_data$sample <- sample
```

```r
kml_data <- kml_data %>%
  mutate(
    x = st_coordinates(.)[, 1],
    y = st_coordinates(.)[, 2]
  )

labels_data <- kml_data %>%
  group_by(Name) %>%
  summarise(
    x = mean(x),
    y = mean(y),
    geometry = st_centroid(st_union(geometry))
  )

# Replace kml_data$Name == "Sakalava_Tsimihety" with "Sakalava/Tsimihety"
kml_data$Name[kml_data$Name == "Sakalava_Tsimihety"] <- "Sakalava/Tsimihety"
# Replace kmldata$Name == "Diego" with "Diego Mixed"
kml_data$Name[kml_data$Name == "Diego"] <- "Diego Mixed"
```

```r
kml_data_single_site <- kml_data %>%
  group_by(Name) %>%
  summarize(geometry = st_union(geometry)) %>%
  st_centroid()
```

Map with population symbols

```r
kml_data <- kml_data[order(kml_data$Name), ]

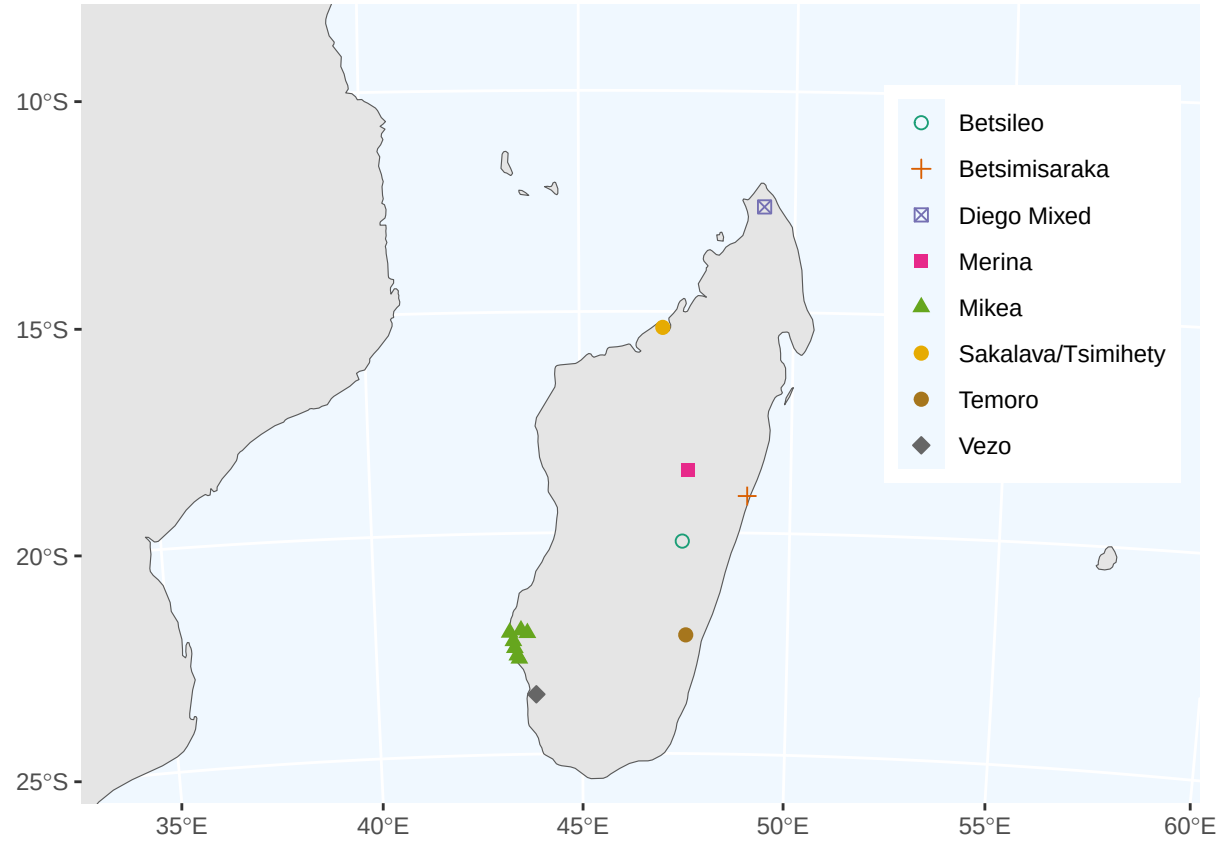
kml_data$shape <- c(1, 3, 7, 15, 17, 17, 17, 17, 17, 17, 17, 19, 21, 23)

main_map <- ggplot() +
  geom_sf(data = madagascar_dissolved) +
  geom_sf(
    data = sf::st_as_sf(kml_data),
    aes(shape = Name, color = Name, fill = Name), size = 2
  ) +
  coord_sf(
    crs = st_crs(29702),
    xlim = c(-1000000, 1800000),
    ylim = c(1, 2000000),
    expand = FALSE
  ) +
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  scale_shape_manual(values = c(1, 3, 7, 15, 17, 19, 21, 23)) +
  labs(shape = NULL) +
  labs(x = NULL, y = NULL, color = NULL, shape = NULL, fill = NULL) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
  guides() +
  theme(
```

```
    legend.position = c(0.85, 0.65),
    legend.background = element_rect(fill = "white")
  )
```

```
## Warning: A numeric ‘legend.position‘ argument in ‘theme()‘ was deprecated in ggplot2
## 3.5.0.
## i Please use the ‘legend.position.inside‘ argument of ‘theme()‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
main_map
```




```
ggsave(paste0(res_dir, "madagascar_map_pca_symbols.jpg"),
  width = 7, height = 5, dpi = 300
)
```

Map with population symbols - single site per pop

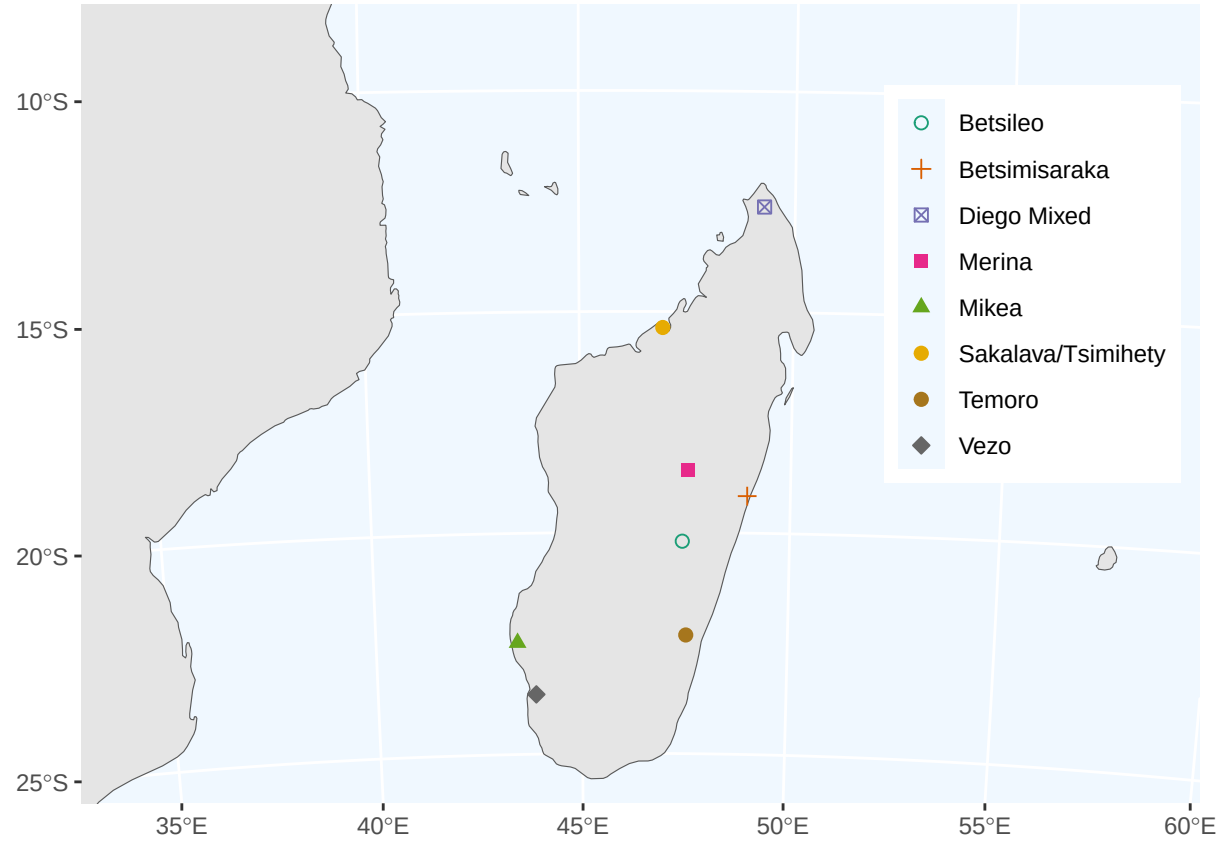
Mikea village locations are averaged into a single point at the centre.

```r
kml_data_single_site <- kml_data_single_site[order(kml_data_single_site$Name), ]

kml_data_single_site$shape <- c(1, 3, 7, 15, 17, 19, 21, 23)

main_map_single_site <- ggplot() +
  geom_sf(data = madagascar_dissolved) +
  geom_sf(
    data = sf::st_as_sf(kml_data_single_site),
    aes(shape = Name, color = Name, fill = Name), size = 2
  ) +
  coord_sf(
    crs = st_crs(29702),
    xlim = c(-1000000, 1800000),
    ylim = c(1, 2000000),
    expand = FALSE
  ) +
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  scale_shape_manual(values = c(1, 3, 7, 15, 17, 19, 21, 23)) +
  labs(shape = NULL) +
  labs(x = NULL, y = NULL, color = NULL, shape = NULL, fill = NULL) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
  guides() +
  theme(
    legend.position = c(0.85, 0.65),
    legend.background = element_rect(fill = "white")
  )

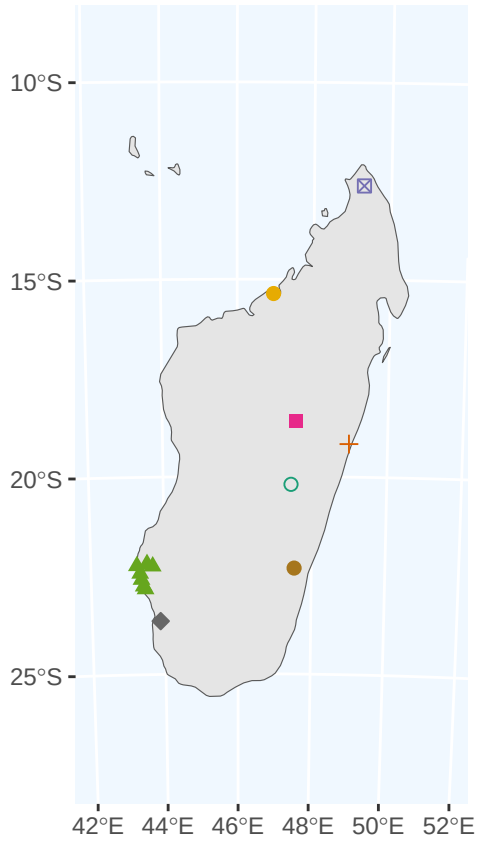
main_map_single_site
```




```
ggsave(paste0(res_dir, "madagascar_map_pca_symbols_single_site.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
p3 <- ggplot() +
  geom_sf(data = madagascar_dissolved) +
  geom_sf(
    data = sf::st_as_sf(kml_data),
    aes(shape = Name, color = Name, fill = Name), size = 2
  ) +
  coord_sf(
    crs = st_crs(29702),
    xlim = c(-100000, 1000000),
    ylim = c(-240000, 2000000),
    expand = FALSE
  ) +
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  scale_shape_manual(values = c(1, 3, 7, 15, 17, 19, 21, 23)) +
  labs(shape = NULL) +
  labs(x = NULL, y = NULL, color = NULL, shape = NULL, fill = NULL) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
  guides() +
  theme(
    legend.position = "none",
    legend.background = element_rect(fill = "white")
```

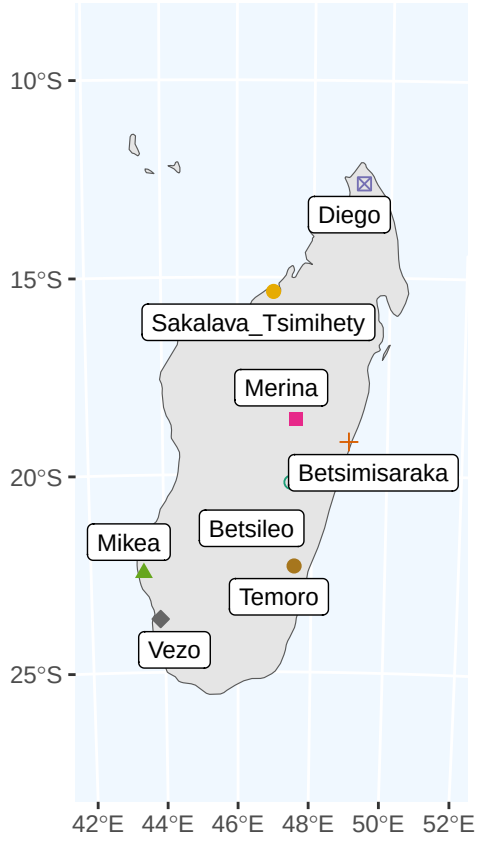
```
  )
p3
```




```
ggsave(paste0(res_dir, "madagascar_map_pca_symbols_slimer.jpg"),
  width = 5, height = 5, dpi = 300
)
```

```
p3 <- ggplot() +
  geom_sf(data = madagascar_dissolved) +
  geom_sf(
    data = sf::st_as_sf(kml_data_single_site),
    aes(shape = Name, color = Name, fill = Name), size = 2
  ) +
  coord_sf(
    crs = st_crs(29702),
    xlim = c(-100000, 1000000),
    ylim = c(-240000, 2000000),
    expand = FALSE
  ) +
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  scale_shape_manual(values = c(1, 3, 7, 15, 17, 19, 21, 23)) +
  labs(shape = NULL) +
  labs(x = NULL, y = NULL, color = NULL, shape = NULL, fill = NULL) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
```

```
  guides() +
  theme(
    legend.position = "none",
    legend.background = element_rect(fill = "white")
  ) +
  geom_label_repel(
    data = labels_data,
    aes(x = x, y = y, label = Name),
    size = 3, segment.color = "transparent"
  )
p3
```




```
ggsave(paste0(res_dir, "madagascar_map_pca_symbols_slimer_single_site.jpg"),
  width = 5, height = 5, dpi = 300
)
```

```
coords_gt_path <-
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_coords.gt"
coord_gt <- gt_load(coords_gt_path)
coord_gt$Latitude <-
  ifelse(coord_gt$population == "Sumba", -9.4, coord_gt$Latitude)
coord_gt$Longitude <-
  ifelse(coord_gt$population == "Sumba", 120.0, coord_gt$Longitude)
coord_gt$Latitude <-
  ifelse(coord_gt$population == "Timor", -9.14, coord_gt$Latitude)
```

```
coord_gt$Longitude <-
  ifelse(coord_gt$population == "Timor", 124.56, coord_gt$Longitude)
# approximate location for Borneo populations
coord_gt$Latitude <-
  ifelse(coord_gt$population == "Borneo", -2, coord_gt$Latitude)
coord_gt$Longitude <-
  ifelse(coord_gt$population == "Borneo", 115.0, coord_gt$Longitude)
coord_gt <- coord_gt %>% filter(!is.na(Latitude))
coord_gt <- gt_add_sf(coord_gt, coords = c("Longitude", "Latitude"))
```

Map of sampling locations

For populations where sampling location metadata was available.

```
new_crs <- suggest_crs(c(-11, 130, -40, 35))

coord_gt <- st_transform(coord_gt, crs = new_crs$proj4)

africa_eurasia <-
  ne_countries(
    continent = c("Africa", "Asia", "Europe", "Oceania"),
    scale = 50, returnclass = "sf"
  )
africa_eurasia <- st_transform(africa_eurasia, crs = new_crs$proj4)

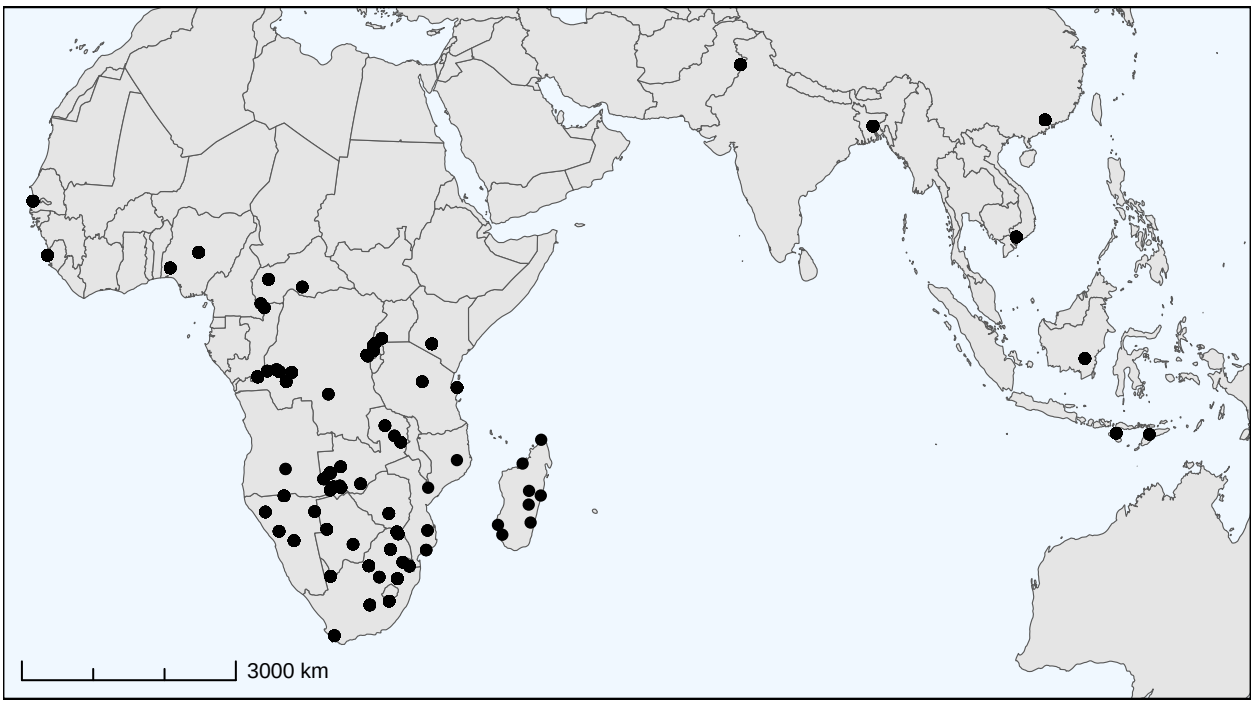
kml_data_single_site <- st_transform(kml_data_single_site, crs = new_crs$proj4)
kml_data_single_site$ours <- c(0, 0, 0, 0, 1, 0, 1, 1)

labels_data <- coord_gt %>%
  group_by(population) %>%
  summarise(
    x = mean(as.numeric(Longitude)),
    y = mean(as.numeric(Latitude)),
    geometry = st_centroid(st_union(geometry))
  )

ggplot() +
  geom_sf(data = africa_eurasia) +
  geom_sf(
    data = coord_gt$geometry
  ) +
  geom_sf(data = kml_data_single_site$geometry) +
  annotation_scale(
    location = "bl", # "bl" = bottom left
    width_hint = 0.2, # Tells R to try and make the bar ~20% of the plot width
    style = "ticks", # Options: "ticks" or "bar"
    unit_category = "metric"
  ) +
  coord_sf(
    xlim = c(-11, 130), ylim = c(-40, 35),
    default_crs = st_crs(4326),
    crs = new_crs$proj4
```

```
  ) +
  theme_minimal() +
  theme_void() +
  theme(
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.5),
    panel.background = element_rect(fill = "#F0F8FF", color = NA)
  )
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
```




```
ggsave(paste0(res_dir, "data_source_map.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
## Scale on map varies by more than 10%, scale bar may be inaccurate
```
